

The film 'The Big Short' covers the 2008 banking crisis (© Paramount Pictures).

FROM THE BIG SHORT TO BIG DATA

HOW BANKS CAN USE DATA SCIENCE TO PREDICT DEFAULTS

Accurate risk prediction is the backbone of modern banking. When models fail the consequences can be catastrophic. How data science and calibration techniques can make credit-risk models more reliable?

"You are saying the rating agencies, the banks and the government are all asleep at the wheel?" Jared Vennett in the movie *The Big Short* just gave his iconic Jenga-tower demonstration to illustrate the vulnerability of America's housing market to defaulting bank loans. The film covers the 2008 banking crisis. The businessmen who just received his pitch are left in disbelief. It is a legendary example of how crucial models are in predicting the likelihood of borrowers defaulting.

Just as in the movie, banks group borrowers into rating buckets, each with a long-run probability of default. These ratings determine how much capital a bank must hold and whether a loan is considered risky or safe. If the

ratings misrepresent risk, as in *The Big Short*, banks may hold too little capital and collapse. Holding too much capital, on the other hand, hurts profitability.

RATING AND CALIBRATION TECHNIQUES

Suppose you are a bank with a model that assigns every borrower a predicted probability of default (PD). There is no single rule for how to translate these PDs into rating grades or how to calibrate them. Different banks do it differently, leading to different capital outcomes.

In his thesis *The impact of rating definition methods and calibrating probabilities of default on capital* Rik Teugels systematically tests combinations of eight rating methods and five calibration techniques to determine which choices truly matter.

Before a bank can trust its model, it must validate it. The predictions should match reality. Credit-risk validation typically examines Expected Loss (EL), Unexpected Loss (UL), and the Brier Score.

- Expected Loss (EL) represents the average amount the bank expects to lose. It is the model's best estimate of the mean outcome, much like a trend line in regression.
- Unexpected Loss (UL) shows how much actual results may fluctuate around that mean. It is a representation of the shocks that require capital buffers.
- The Brier Score compares predicted probabilities with real defaults, measuring how well-calibrated the model is. This score was used as a means of validating the pipeline.

Each rating bucket receives a probability of default meant to match either historical defaults or regulatory expectations. However, model predictions often deviate from those targets, requiring a calibration step to realign predictions with reality. Calibration adjusts the probability scale so that predicted and observed default rates coincide.

CALIBRATION MAKES THE DIFFERENCE

The thesis's central finding is clear: calibration matters far more than how borrowers are bucketed. The rating step only ranks borrowers by relative risk, while calibration ensures those probabilities accurately reflect the real world. A flexible calibration method can even offset small differences in binning strategies.

Among the techniques studied, isotonic regression and historical-default calibration achieved the best balance between accuracy and stability. The former smooths the relationship between predicted and observed defaults, while the latter directly assigns each grade its historical default rate. Both are transparent, data-driven, and free from heavy economic assumptions.

A clear message for both data scientists and financial risk managers: get calibration right.

In conclusion, Rik Teugels' research delivers a clear message for both data scientists and financial risk managers: get calibration right. Grouping borrowers is secondary; calibration transforms statistical predictions into trustworthy probabilities. Done well, it enables banks to determine realistic capital requirements – and avoid the next *Big Short*.



Thesis title: The impact of rating definition methods and calibrating probabilities of default on capital

Author: Rik Teugels

Institution: KU Leuven (Master of Business and Information Systems Engineering)

Why this thesis stands out

This thesis offers empirical proof of how calibration decisions can reshape capital requirements. It turns an abstract data-science concept into a practical tool for risk governance.

Key takeaways

- Calibration drives accuracy: it determines whether default probabilities reflect reality.
- Binning methods matter less: the order of risk remains, but calibration corrects the scale.
- Isotonic regression and historical-default calibration offer the best balance of precision and stability.
- Proper calibration ensures credible risk models and reliable capital buffers.

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